

An AI-Assisted Pipeline for Puzzle Hunt Creation: Design, Empirical Characterization, and Leverage Points

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Abstract

Generating a single creative artifact with AI — a poem, a puzzle, a game level — is now routine. Generating a *coherent collection* of such artifacts — a puzzle hunt, with a shared thematic world, a set of interlocking puzzles, an answer-confirming meta-puzzle, and a finished delivery package — is a qualitatively different problem. It requires coordinated decisions across scope, authorship, structure, and delivery, and it exposes a gap that single-shot generation cannot close: without process structure, AI output quality degrades badly across a multi-artifact production task.

We address this gap with a structured, 11-stage generative pipeline for puzzle hunt creation. Each stage produces a defined set of artifacts consumed by subsequent stages; two stages function as formal review gates — Stage 3 (puzzle pool ranking) and Stage 7 (editorial review) — where an AI expert panel evaluates candidate puzzles against a published rubric and issues pass/revise/cut verdicts. The pipeline is implemented as a library of 13 reusable CLAUDE.md skill prompts, requiring no per-scenario code changes.

We evaluate the pipeline across five fully-tested hunt scenarios spanning five distinct domains (a real-time strategy game, noir detective fiction, science fiction role-playing, music, and print puzzles) and three delivery formats (physical dossier, digital website, print-and-play). Across 52 puzzles, the pipeline achieves a 90.4% first-pass pass rate (47/52) (against the pipeline’s AI panel rubric, not an external community standard). The five failures are not uniformly distributed: three of the five occur at Stage 8 (testing and integration), downstream of both review gates, and zero failures are attributable to Stage 3. This distribution confirms that the gates are functioning — they surface quality problems before authoring investment is made — and that the dominant failure mode is post-gate integration, not generation quality per se.

We contribute: (1) a precise specification of the 11-stage pipeline with gate conditions; (2) cross-scenario empirical characterization across 52 puzzles and five domains; (3) a failure analysis identifying Stage 8 as the

concentrated failure point; (4) identification of Stages 3 and 7 as highest-leverage interventions; and (5) an open pipeline specification reproducible from a public git repository (142 commits, two active days) with no proprietary dependencies. The first scenario completed in 3h 17m elapsed wall-clock time; active human effort is not separately measured.

1 Introduction

A puzzle hunt is not a puzzle. This distinction, obvious to the puzzle hunt community, carries real technical weight for anyone attempting to generate one with AI. A single puzzle — a crossword, a logic grid, a cryptic clue — is a contained artifact with a well-defined quality criterion: it either works mechanically or it does not. A puzzle hunt is a *multi-artifact creative system*: a set of thematically unified puzzles, each with a confirmed final answer, feeding into a meta-puzzle whose extraction depends on the feeders’ specific answers, all wrapped in an editorial voice, a narrative world, and a delivery package formatted for a specific medium (a website, a physical dossier, a print-and-play booklet). The quality bar for a hunt is not just “does each puzzle work?” but “does the collection cohere? does the meta close? does the delivery hold together?”

This problem class — multi-artifact creative system generation — is not well addressed by existing AI generation techniques. Single-shot prompting produces variable output uncorrelated across artifacts; it has no mechanism for the kind of global constraint propagation that puzzle hunt design requires. Agent-loop approaches [?, ?] introduce iteration but do not inherently impose the structured handoffs that keep downstream artifacts consistent with upstream decisions. Procedural content generation (PCG) research has addressed related problems in level generation [?, ?] and narrative [?, ?], but these systems typically target single-artifact or fixed-grammar outputs, not a heterogeneous pipeline of interdependent creative artifacts.

The core difficulty is *coordination across decisions*. A puzzle hunt requires that the world-building done at Stage 1 constrain the puzzle types admitted at Stage 3, that the puzzle answers selected at Stage 4 remain stable for the meta-puzzle designed at Stage 5, and that the delivery format chosen at Stage 1 be honored by the assets produced at Stage 9. Without explicit structure, AI generation at the scale of a hunt degenerates: the world drifts, the meta fails to close, puzzles assume facts the world has not established, and delivery assets reference puzzles that have been cut. Coherence collapse is the default outcome of unconstrained multi-artifact generation.

Our Approach

We address this problem with a structured, 11-stage AI-assisted pipeline in which each stage produces a defined set of artifacts consumed by subsequent stages, and two stages function as formal review gates with explicit pass criteria. The pipeline is not an agent loop and not a single prompt chain. It is a

decomposition of the full puzzle hunt production task into stages that can be executed, reviewed, and retried independently, with structured artifacts (markdown documents following defined templates) serving as the sole interface between stages. Stage 6 (Authoring) in particular operates under dense human constraints — a fully-specified brief, a locked world, and a committed answer word — so the pipeline is better characterized as AI-assisted execution under specification than as open-ended AI creativity. A *world lock* mechanism — a canonical world document treated as read-only after Stage 1 — prevents coherence drift across the pipeline by ensuring that all downstream generation is grounded in a single approved source of truth [?].

Two stages are designated review gates:

- **Stage 3 (Puzzle Pool Ranking):** An AI expert panel evaluates a pool of puzzle candidates against a published rubric and assigns numeric scores. Only candidates meeting the score threshold (or approved by human override) proceed to authoring. This gate is deliberately early: catching weak candidates before authoring investment is made is cheap; catching them after authoring is expensive.
- **Stage 7 (Editorial Review):** After authoring, each puzzle is submitted to a second AI panel for editorial review. The panel issues a pass/revise/cut verdict with specific, actionable feedback. Only puzzles passing Stage 7 proceed to integration. This gate prevents broken puzzles from entering the meta-puzzle and delivery package.

The pipeline is implemented as a library of 13 CLAUDE.md skill prompts — persistent markdown instruction files that configure an AI assistant to execute a specific pipeline stage. The library runs without per-scenario code changes: the same skill prompts ran all five evaluated scenarios, with only per-scenario world and scope documents varying.

Key Empirical Questions

Given a structured pipeline with explicit gate conditions, three empirical questions become tractable:

1. **Does stage structure matter for output quality?** The 90.4% first-pass pass rate across 52 puzzles provides an aggregate quality signal; failure distribution across stages provides the structural explanation.
2. **Which stages are the highest-leverage intervention points?** If failures are uniformly distributed across stages, then gate placement is arbitrary. If failures concentrate at specific stages, that concentration identifies where pipeline investment has the highest return.
3. **Is the pipeline transferable across themes and settings?** If pipeline quality holds only for the theme it was developed in, it is a one-off system, not a general contribution. If it transfers to new themes and settings without code changes, the pipeline specification itself is the contribution.

We ground these questions in five fully-executed hunt scenarios spanning five distinct thematic domains (a real-time strategy game, noir detective fiction, science fiction role-playing, music, and print puzzles) and three delivery formats. The first scenario (Age of Empires “Wololo”) completed Stages 1 through 10 in 3 hours and 17 minutes (elapsed wall-clock time; active human effort not separately measured) as measured by git commit timestamps; subsequent scenarios ran in overlapping sessions, with the full pipeline executed independently for each. Across all five scenarios, 52 puzzles were authored, reviewed, and tested.

Contributions

This paper makes the following contributions:

1. **Pipeline specification.** We specify an 11-stage AI-assisted puzzle hunt pipeline with defined input/output artifacts per stage, explicit gate conditions at Stages 3 and 7, and an autonomy gradient characterizing the degree of AI versus human agency at each stage.
2. **Cross-scenario empirical characterization.** Across five hunt scenarios and 52 puzzles spanning five domains and three delivery formats, the pipeline achieves a 90.4% first-pass pass rate (47/52). We provide per-scenario and aggregate pass rate breakdowns. The first fully-executed scenario completed in 3h 17m (elapsed) from scope to ship-ready artifacts.
3. **Failure analysis.** Five puzzles failed first-pass review. Three of the five failures occurred at Stage 8 (testing and integration), downstream of both review gates; zero failures were attributable to Stage 3 (pool gate). We categorize failure types (mechanism failure, answer coordination failure, construction error, delivery format gap) and map each to its earliest detectable stage.
4. **Identification of highest-leverage stages.** The failure distribution confirms Stage 3 and Stage 7 as the pipeline’s highest-leverage interventions. Stage 3 gates the investment of authoring work: failing here is cheap. Stage 7 gates integration: failing here wastes authoring and blocks meta closure. The concentration of failures *after* Stage 7 (at Stage 8), and zero failures at Stage 3, confirms that both gates are functioning as intended.
5. **Open pipeline specification.** The full pipeline is specified in a public git repository: 142 commits over two calendar days, a skill library that grew from 0 to 13 skills, and a reviewer profile library that grew from 12 to 29 profiles. The repository contains all scenario artifacts (scope documents, puzzle files, editorial review reports) sufficient to replicate or extend the pipeline to new domains without proprietary dependencies.

Paper Organization

Section 2 situates the pipeline within prior work on PCG, AI-assisted creative systems, and review-gate patterns in creative production. Section 3 describes the pipeline in full, including stage-by-stage artifact specifications, gate conditions, and the implementation as a CLAUDE.md skill library. Section 4 presents the empirical characterization: scenario overview, pass rate analysis, stage timing, failure analysis, and cross-theme transferability. Section ?? discusses the implications of the failure distribution, the role of the world lock mechanism, and limitations of the current evaluation. Section ?? summarizes contributions and identifies open questions.

2 Related Work

2.1 AI-Assisted Creative Systems

Research on AI-assisted creative generation has overwhelmingly focused on single-artifact outputs: a poem, an image, a game level, a short narrative. Systems in this tradition treat generation as a single forward pass from prompt to artifact, with quality evaluated against criteria defined for that artifact class in isolation. Roemmele and Gordon [?] demonstrate neural story continuation at the level of individual narrative segments; Summerville et al. [?] survey level generation systems that produce one playable map at a time. The AI co-authorship literature [?] similarly takes the individual artifact—a text, a scene, a game element—as the unit of generation and the unit of evaluation.

Puzzle hunts, tabletop campaigns, and other multi-artifact creative systems present a structurally different problem. The unit of generation is not a single artifact but a *coherent collection* of artifacts that must satisfy both per-artifact quality criteria and cross-artifact consistency constraints: thematic unity, answer set compatibility, difficulty arc, meta-puzzle closure. Generating such systems requires coordinated decisions across multiple independent generation events, separated in time and context. Park et al.’s generative agents [?] demonstrate that LLMs can sustain coherent behavior across multi-turn interactions through memory and identity representations, but their setting is social simulation rather than structured creative production—the agents’ outputs are not artifacts consumed as inputs by downstream generation stages.

The key gap is that no prior work addresses coherent *multi-artifact system generation* across multiple independent generation runs, where each run’s output becomes a structured constraint on subsequent runs. Single-artifact systems cannot be trivially composed into multi-artifact pipelines: coherence drift accumulates across generations unless explicitly suppressed by a cross-stage consistency mechanism. Our pipeline addresses this gap through stage-gated generation with a canonical world document that functions as a shared read-only constraint across all stages.

2.2 Procedural Content Generation with Constraints

Procedural content generation (PCG) provides the nearest prior framework for understanding structured creative pipelines. Shaker et al. [?] survey the field broadly, distinguishing generate-and-test approaches, constructive algorithms, and search-based methods. Smith and Whitehead [?] develop Tanagra, a mixed-initiative level design system that treats design constraints as real-time feedback to the generator, demonstrating that constraint propagation across design decisions improves both output quality and designer agency. The PCG-via-search paradigm in particular operationalizes quality as a fitness function and treats generation as search over an artifact space bounded by explicit constraints.

Our pipeline is constraint-based in the same spirit, but the constraint structure is sequential across stages rather than concurrent within a single generation pass. Each stage’s output artifacts become binding constraints on subsequent stages: the world document (Stage 1) constrains all thematic content generated in later stages; the puzzle pool ranking (Stage 3) constrains which puzzle designs proceed to authoring; the feeder answers committed at authoring time (Stage 6) constrain meta-puzzle design (Stage 5). This staged constraint propagation differs from mixed-initiative PCG [?] in an important respect: the constraints are not relaxed or negotiated during generation, but locked at explicit gate conditions. The world lock mechanism is not a soft preference but a hard read-only boundary—downstream stages are explicitly prohibited from modifying it. This design choice prioritizes coherence over flexibility, accepting that late-stage discoveries cannot propagate back to world-level decisions.

The review-gate pattern that enforces these constraints—requiring an explicit pass verdict from an evaluation stage before proceeding to the next generation stage—has no direct precedent in the PCG literature, where quality filters are typically applied post-hoc or embedded as fitness functions within a generative loop. Our gates are pre-hoc: they prevent low-quality artifacts from entering the pipeline segment where downstream authoring investment would be wasted on them.

2.3 Puzzle Design and Quality

The practitioner community has developed explicit quality standards for puzzle design that are more codified than most creative domains. The Riven Standard, articulated by Rand Miller in reflecting on the *Myst* sequel, holds that a well-designed puzzle is one whose mechanic is inseparable from its setting: the puzzle should *be* what the domain does, not merely occur within it [?]. Snyder [?] enumerates craftsmanship criteria for puzzle construction that translate across types: every element should earn its presence, solutions should be uniquely determined, and the solver’s confirmation should be unambiguous. Competitive puzzle construction standards from the World Puzzle Championship and U.S. Puzzle Championship instantiate these principles as elimination criteria, providing a community-validated quality bar that is unusually explicit relative to other creative fields.

Puzzle hunts introduce quality dimensions not present in standalone puzzle evaluation. A puzzle hunt is not a collection of independent puzzles but a structured system: puzzles must yield answer words that interact correctly with the meta-puzzle, maintain consistent difficulty calibration across rounds, and sustain thematic coherence with the hunt’s fictional premise. The meta-puzzle itself introduces a structural correctness criterion—it must be solvable from the feeder answers it receives, and it must close cleanly—that has no analog in standalone puzzle design. Answer confirmation, a dimension in our Stage 7 rubric, is unique to the hunt context: it asks whether solving the puzzle produces an unambiguous signal of correctness, enabling teams to verify their answer without adjudication.

The AI expert panel review system deployed at Stages 3 and 7 of our pipeline operationalizes these practitioner criteria as a six-dimension rubric (Clarity, Solvability, Elegance, Reading Reward, Fun, Confirmation) with empirically grounded thresholds. The design and calibration of this system is the subject of a companion paper [?]; the present paper treats it as a component and characterizes its behavior as a pipeline gate across six independent hunt scenarios.

2.4 AI Pipelines for Complex Tasks

The agent orchestration literature has proposed several frameworks for decomposing complex tasks across multiple LLM invocations. Park et al. [?] demonstrate that generative agents with persistent memory and identity can sustain coherent behavior across multi-turn social simulations. Nakajima’s BabyAGI [?] and Richards’s AutoGPT [?] popularized autonomous agent loop architectures in which a planning agent generates, prioritizes, and executes subtasks without human intervention between steps. More structured orchestration frameworks such as AutoGen [?] and LangChain [?] provide scaffolding for multi-agent conversation and sequential tool use, enabling pipelines where agent outputs are routed as inputs to downstream agents.

Our pipeline differs from this tradition in a design decision that is deliberate rather than incidental: we do *not* use autonomous agent loops. Each of the eleven stages is human-triggerable and independently reviewable; no stage executes automatically as a consequence of the previous stage completing. This design makes the pipeline slower than an autonomous loop architecture, but it addresses a failure mode that autonomous loops inherit: unchecked error propagation. In an autonomous loop, a generation error at Stage 3 that escapes detection will cause downstream stages to invest authoring time on a flawed artifact, and the error may not be visible until Stage 7 or Stage 10. In our pipeline, a human reviews the output of each stage before triggering the next, and formal gate conditions at Stages 3 and 7 require an explicit pass verdict before downstream work begins.

This *review-gate pattern* relates to two bodies of practice outside the AI literature. In traditional publishing, editorial pipelines impose stage gates between acquisitions, developmental editing, copy editing, and production, with each gate requiring editorial approval before investment in the next stage [?]. In

software engineering, quality gates in continuous integration pipelines enforce structural analogs: automated and human checks must pass before a change is promoted to the next environment. Our pipeline adapts this logic to AI content generation, treating review stages not as overhead but as the primary mechanism by which generation quality is enforced. The empirical question this paper addresses is where in an eleven-stage pipeline these gates add the most value—and the answer, across six independent hunt scenarios, is that Stages 3 and 7 account for the majority of quality control, with failure rates at other stages negligible by comparison.

3 System Description

This section describes the 11-stage pipeline as a designed system: its stage structure, artifact dependencies, gate conditions, AI autonomy model, and implementation substrate. The system contribution is the pipeline specification itself — a reusable process architecture that can be instantiated across hunt domains without modification.

3.1 Pipeline Overview

The pipeline consists of eleven sequential stages, each producing a structured artifact that serves as the primary input to the next stage. Figure 1 provides a schematic of the full stage sequence; the two review gates (Stages 3 and 7) are marked as decision points. Table 1 provides the full stage reference.

Table 1: Pipeline stage reference. AI autonomy levels: **FA** = fully AI-autonomous; **AP** = AI-primary, human-confirms; **PD** = AI-panel-driven (review gate); **HP** = human-primary. Gate condition applies only to stages with a formal pass/fail threshold.

#	Stage	Primary Input	Primary Output	Gate Condition	AI Role
1	Scope	Hunt brief (prose)	SCOPE.md, CLAUDE.md	Human approval	AP
2	World Build	SCOPE.md	world/, LOCKED.md	—	AP
3	Puzzle Pool	World data, ROUNDS.md	PUZZLE-POOL.md (scored)	Score $\geq 33/45$	PD
4	Assignment	PUZZLE-POOL.md	Per-puzzle ASSIGNMENT.md	—	FA
5	Meta Design	Feeder answers	META-DESIGN.md	Human verification	AP
6	Authoring	Per-puzzle ASSIGNMENT.md	Puzzle files (one per puzzle)	Score $\geq 22/30$	FA
7	Editorial	Puzzle files	Per-puzzle REVIEW.md	Pass/revise/cut	PD
8	Integration	Editorial-passed puzzles	PUZZLES.md integration checklist	—	FA
9	Delivery Build	Integrated puzzle set	Site, print, or props package	—	AP
10	Platform Test	Delivery package	Solver simulation report	—	FA
11	Polish	Simulation report	Final artifacts, hint system	—	HP

The pipeline is *artifact-driven*: each stage’s output is a version-controlled markdown document, and subsequent stages treat those documents as read-only inputs. This constraint enforces unidirectional information flow and prevents

coherence drift, which would otherwise occur when late-stage authoring decisions propagate backwards to modify scope or world documents. A dedicated *world lock* enforces this at the content level: Stage 2 produces a `LOCKED.md` canon declaration, and all subsequent stages are prohibited from introducing facts not present in the locked world files. Every factual claim in every puzzle must be traceable to a specific entry in the `world/` directory. This mechanism was validated empirically: across nine hunt scenarios, no post-lock canon violations were detected in Stage 7 editorial review, compared to a baseline of multiple coherence errors per scenario in early pilot runs conducted without the lock mechanism.

Stage 1 (Scope) establishes the hunt’s identity: domain, audience, scale, tone, delivery format, and thematic frame. The primary output is a `SCOPE.md` document and a per-scenario `CLAUDE.md` status tracker. The status tracker is the pipeline’s runtime state: each stage updates it on completion, and the `/hunt resume` skill reads it to recover from session interruptions without duplicating completed work.

Stage 2 (World Build) populates the content library from which puzzles draw their facts. For fictional domains (Ironfall’s 1993 JRPG universe, Dead Reckoning’s starship systems), the world is synthesized and populated into structured data tables. For real-world domains (Age of Empires, Wavelength’s pop music corpus), the world stage collects, verifies, and canonicalizes factual data. The `/hunt world` skill organizes this data into individual system files (bestiary, items, locations, characters, and so on) and then issues the canon lock.

Stage 3 (Puzzle Pool) generates a pool of candidates — typically three times the number of puzzles needed for the final hunt — and subjects them to AI panel ranking (see Section 3.2). The scored pool is the primary input to Stage 4 (Assignment), which allocates the highest-scoring candidates to puzzle slots and coordinates all feeder answer words to ensure meta compatibility.

Stage 5 (Meta Design) is intentionally ordered before Stage 6 (Authoring): the meta puzzle’s extraction mechanism determines which feeder answer words are required, and all feeder puzzles must be briefed with their target answer words before authoring begins. This ordering prevents the most common failure mode in AI-generated hunts — meta answers that do not match the feeder set — by establishing answer contracts before any puzzle content is written.

Stage 6 (Authoring) writes puzzle files from the coordinated briefs. In the multi-scenario runs described in Section 4, Stage 6 was executed as parallel AI agent launches, with one or more independent agents per puzzle, running simultaneously without shared state. Each agent reads its assignment brief, the relevant world data files, and the design principles document, then produces a complete puzzle page. No human editing occurs between brief and authored output.

Stages 7 through 11 constitute the review-to-ship pipeline: editorial review (with gate), cross-puzzle integration verification, delivery format construction, platform simulation, and final polish. Stage 11 is the only stage coded as human-primary (see Section 3.3): it encompasses hint writing, edge case detection, and final answer encoding — tasks that require human judgment about solver

experience that AI simulation does not reliably approximate.

3.2 Gate Conditions

The pipeline contains two primary review gates: Stage 3 (pool ranking) and Stage 7 (editorial review). Both are implemented using the AI expert panel system described in [?]. A secondary gate operates within Stage 6 (authoring): each puzzle must pass an automated quality threshold before proceeding to editorial.

Stage 3: Pool Ranking Gate. The puzzle pool gate filters candidates before the investment of authoring work. The panel evaluates each candidate against the seven-dimension weighted rubric introduced in [?]: Clarity ($\times 1$), Solvability ($\times 1$), Elegance ($\times 2$), Reading Reward ($\times 2$), Fun ($\times 1$), Confirmation ($\times 1$), and Riven Standard ($\times 1$), yielding a maximum score of 45. At Stage 3, candidates are evaluated on their *brief* — the one-to-two paragraph specification of the puzzle concept, mechanism, and extraction — rather than on authored content. The pass threshold is 33/45 (73%), matching the threshold calibrated in the rubric validation experiment of [?] against three quality tiers of published puzzle hunt puzzles.

Candidates scoring below 33 are not assigned. Human override is permitted: a hunt designer may promote a candidate below threshold if the brief’s weakness is judged recoverable in authoring, or suppress a high-scoring candidate for editorial reasons (e.g., domain overlap with another selected puzzle). In practice, across six scenarios, human override occurred in fewer than 5% of pool decisions; the panel’s ranking was treated as authoritative in the large majority of cases.

The Stage 3 gate is designed to fail cheaply. At the point of pool evaluation, the only work invested is the candidate brief (typically 200–400 words). Failing a candidate at Stage 3 therefore wastes at most the brief — not the full authoring effort, which for a complete puzzle is an order of magnitude larger. The gate is explicitly calibrated to have high recall rather than high precision: it is better to promote a borderline candidate that fails at Stage 7 than to suppress a strong candidate that a brief undersells.

Stage 7: Editorial Review Gate. The editorial gate operates on fully authored puzzles and produces a verdict in one of three categories: PASS (puzzle proceeds to integration), REVISE (puzzle returns to Stage 6 with a specific revision brief), or CUT (puzzle is removed from the hunt). The panel evaluates authored content against the same rubric used in Stage 3, augmented by three additional editorial checks: extraction chain verification (character-by-character), voice consistency audit (does the puzzle match the hunt’s established narrative voice), and world-lock compliance (does every factual claim trace to a locked world file).

A REVISE verdict triggers a targeted revision pass: the authoring agent receives the review feedback and rewrites the specific extraction mechanism, clueing logic, or presentation element flagged by the panel. In the scenarios reported

here, revisions were executed in a single pass in all cases where revision was required; no puzzle required more than one revision cycle. A CUT verdict was reserved for puzzles with fundamental mechanism failures that would require redesign from the brief, which was judged more efficient than revision.

The Stage 7 gate is the pipeline’s primary quality control mechanism. Unlike the Stage 3 gate, which screens design concepts, Stage 7 screens executed construction. The failure modes it catches are different: not “is this concept worth pursuing” but “does this puzzle actually work as built.” The empirical data (Section 4) shows that Stage 7 catches failures that Stage 3 does not predict: concept-level scores are not reliable predictors of construction-level failures, which are typically mechanism errors (extraction chains that do not resolve) or answer coordination errors (feeder answers that do not match meta requirements).

Why These Two Gates. The two-gate design reflects a deliberate cost-staging argument. Creative pipeline failures are not equally expensive at all stages; the cost of failure is proportional to the downstream investment that must be discarded. Stage 3 failures are cheap (brief only); Stage 7 failures are moderate (authoring effort only, before integration); failures that reach Stage 8 or beyond are expensive (integration, delivery build, and test work is lost or must be redone). Placing review gates at Stages 3 and 7 maximizes the expected quality of work that reaches integration while minimizing total evaluation overhead. No gate is placed between Stage 7 and Stage 11 because the editorial and integration stages are jointly sufficient to catch remaining failures — which, empirically, are concentrated in delivery format issues rather than puzzle construction failures.

3.3 AI Roles by Stage

Each stage in the pipeline is coded on a four-level autonomy gradient: **FA** (fully AI-autonomous), **AP** (AI-primary, human confirms), **PD** (AI-panel-driven, with human deciding on revisions), and **HP** (human-primary). Figure ?? illustrates the gradient across the full pipeline. The coding is based on the empirical record of how each stage was executed across all scenarios in this study.

Stages 2, 4, 6, 8, and 10 are fully AI-autonomous in practice. Stage 2 (World Build) generates and populates content tables from domain specifications with no human editing of individual entries. Stage 4 (Assignment) translates pool rankings to puzzle slot assignments and generates all briefs without human intervention. Stage 8 (Integration) runs the integration checklist — answer verification, cross-reference checking, round structure validation — autonomously. Stage 10 (Platform Test) executes a solver simulation against the delivery package, with each simulated solver personality working through the hunt independently and logging failures.

Stage 6 (Authoring) warrants specific discussion as the most autonomous stage in the pipeline. Puzzles are authored by independent AI agents launched in parallel, one per puzzle (or one per module in large hunts), with no human review

occurring between brief and authored output. The agents share no state; they communicate only through the coordinated answer words established in Stage 5. In the largest scenario in this study (Cosmere, 36 feeder puzzles across four modules), 36 puzzles were authored in parallel across a single session without human editorial intervention until Stage 7. This represents a qualitative shift from human-assisted authoring: the human’s role in Stage 6 is to launch agents and wait for completion, not to review or approve individual puzzle content. Stage 6 autonomy is feasible precisely because Stage 5 has pre-coordinated all answer words and Stage 7 will catch authoring failures.

Stage 1 (Scope) and Stage 5 (Meta Design) are coded AP: AI generates the scope document or meta mechanism, and the human confirms before proceeding. These confirmations are lightweight — typically a review of the SCOPE.md document or a meta extraction trace — but they are genuine decision points at which the human can redirect the hunt’s identity before downstream work commits to a particular direction.

Stages 3 and 7 are panel-driven. The AI panel produces scored rankings (Stage 3) or pass/revise/cut verdicts (Stage 7), but the human decides on override and revision scope. In practice, these stages are AI-primary in execution and human-primary in decision authority; the panel output is treated as a strong recommendation, not an automatic action.

Stage 11 (Polish) is the pipeline’s most human-intensive stage. Hint writing requires calibrating the difficulty of each hint tier to the expected solver population — a judgment about cognitive load that AI simulation does not reliably approximate. Edge case detection requires the human to reason about degenerate solver paths that the platform test simulation may not have exercised. Final answer encoding and security checks (ensuring no plaintext answers appear in any distributed file) are procedurally defined but require human sign-off before the hunt ships. Stage 11 is also where the hunt designer’s creative voice is most directly expressed: the final pass shapes the solver’s experience in ways that are not captured by the quality rubric.

3.4 Implementation

The pipeline is implemented as a *CLAUDE.md skill library*: a collection of persistent markdown documents, each encoding a named skill as a structured prompt with defined inputs, defined outputs, and explicit instructions for interacting with the scenario’s artifact directory. Skills are installed into Claude Code’s persistent skill store (the `~/claude/skills/` directory) and invoked by slash command (e.g., `/hunt world`, `/puzzle P01 author`). No bespoke code is written for any scenario; the entire pipeline executes through natural language skill invocations against version-controlled artifact files.

Skills are organized into two namespaces. The `/hunt` namespace contains hunt-level administrative skills run by the hunt director: `plan` (the full pipeline walkthrough), `world` (world build and lock), `review` (panel review invocation), `meta` (meta design and verification), `edit` (editorial review pass), `resume` (crash recovery), `print` (print package generation), `site` (website scaffold and puzzle

pages), **props** (physical prop logistics), **publish** (solver-facing distribution package), **script** (game architecture script for interactive delivery), **status** (pipeline dashboard), and **profile** (reviewer profile management). The **/puzzle** namespace contains puzzle-level skills used by authoring agents: **author** (write puzzle from brief), **test** (blind test against quality threshold), and **check** (design principles audit).

The skill library grew from 0 to 13 skills over the two-day study period (February 27–28, 2026), with skills added reactively in response to pipeline needs discovered during scenario execution. Table ?? in Section 4 provides the full growth timeline. This organic development trajectory is itself an empirical finding: the 13 skills that emerged are those that were actually needed to run the pipeline end-to-end, not a pre-specified design. Skills added late in the study (notably **/hunt script** and **/hunt publish**) address delivery and distribution tasks that only become visible as a bottleneck when multiple scenarios reach Stage 9 simultaneously.

Each scenario’s per-scenario **CLAUDE.md** references the toolkit skill library and specifies only three per-scenario parameters: domain (the content universe or real-world topic), audience (target solver population and experience level), and scale (puzzle count, round structure, and delivery format). No per-scenario modifications to skill prompts were required across any of the nine scenarios in this study. The same **/hunt world lock**, **/puzzle P01 author**, and **/hunt edit** commands that ran the Age of Empires scenario in February also ran the Cosmere scenario with 36 puzzles and four meta layers, without any prompt engineering between scenarios. This cross-theme invariance is the primary evidence for the pipeline’s transferability claim.

The **/hunt resume** skill deserves specific mention as an infrastructure component. Claude Code sessions are stateful but not persistent: a session crash loses the in-context working memory of all stages executed in that session. The resume skill recovers from crashes by reading the per-scenario **CLAUDE.md** status tracker (updated at every stage boundary), identifying the first incomplete stage, and regenerating the necessary context from the artifact files on disk before continuing. In Stage 6, resume behavior is puzzle-aware: it checks each puzzle individually and re-authors only puzzles that do not yet have a completed file, skipping puzzles that were successfully authored before the crash. Across the multi-session, multi-scenario runs of this study, the resume skill was invoked at least once in the majority of scenarios, confirming that crash recovery is a practical operational requirement rather than an edge case.

4 Empirical Characterization

We evaluated the pipeline across five fully-tested hunt scenarios, spanning four distinct content domains (themes and settings), three delivery formats, and a scale range of 6 to 19 total puzzles. This section reports aggregate pass rates, per-stage timing, failure analysis, and evidence for cross-theme transferability. All measurements are derived from commit timestamps and scenario artifact

[Figure 1: Pipeline diagram — 11 stages in sequence, with gate condition markers at Stages 3 and 7, input/output artifact labels per stage, and autonomy gradient coding. TikZ source in `figures/fig1-pipeline.tex`.]

Figure 1: The 11-stage AI-assisted puzzle hunt pipeline. Review gates (Stages 3 and 7) are the primary quality control interventions. The autonomy gradient shifts from AI-primary in the middle stages (authoring, integration, testing) to human-primary at the boundaries (scope and polish).

files in the repository.

4.1 Scenario Overview

Table 2 summarizes the five scenarios that completed the full 11-stage pipeline with verified editorial review. A sixth scenario (Boardgames “Game Night”) reached Stage 6 before being suspended as a partial workflow test; it is excluded from aggregate pass-rate calculations but its single Stage 6 failure is included in the failure analysis (§4.4).

Table 2: Summary of the five fully-tested pipeline scenarios. Pass rate is reported pre- and post-redesign where redesigns occurred. “Candidates” is the size of the Stage 3 puzzle pool before selection; “Final count” is the number of puzzles shipped.

Scenario	Domain	Scale	Delivery	Meta answer	Pool	Final	Pass rate
Age of Empires	Real game (RTS)	5+1	Print/digital	WOLOLO	15	6	83%→100%
Grand Larceny	Noir fiction	4+1	Physical dossier	SLIT	—	6	100%
Ironfall	Sci-fi RPG (fiction)	10+3	Website	Konami Code	18	13	100% [†]
Wavelength	Music / thematic	6+1	Website + prop	FADING	—	7	100%
Dead Reckoning	Sci-fi investigation	15+4	Website	CALIBRATE	45	19	100% (v)
Total						51	

[†] 100% post-redesign; 2 puzzles (P01, P02) required extraction redesign at Stage 8.

“—” indicates pool candidates selected within the same session without a discrete count artifact.

The five scenarios collectively span four content domains (real-game knowledge, original noir fiction, original sci-fi fiction, and music), three delivery formats (print/digital, physical dossier, and interactive website), and a fourfold scale range (6 to 19 final puzzles). All five ran against the same skill library and reviewer profile set without per-scenario modifications.

4.2 Pass Rate Analysis

Across the five fully-tested scenarios we track 52 total puzzles (combining the Dead Reckoning verified count with the four live-tested scenarios). Of these, 47 passed first-pass editorial review, yielding an overall first-pass pass rate of **90.4%** (47/52). All 5 failures were subsequently recovered through redesign or delivery remediation, bringing the final ship rate to 100%. These figures measure pass rates against the pipeline’s author-constructed AI panel rubric; they do not represent external validation against solver enjoyment or community quality standards.

Figure ?? breaks down the pass/redesign/cut count per scenario. The four scenarios with live solver simulation (Age of Empires, Grand Larceny, Ironfall, Wavelength) total 32 puzzles, of which 29 passed first-pass (90.6%); 3 required redesign. Dead Reckoning contributes 19 additional verified puzzles at 100%.

Three observations follow from this distribution. First, pass rate does not degrade with scale: the largest scenario (Dead Reckoning, 19 puzzles) achieved 100% first-pass, while the smallest (Age of Empires, 6 puzzles) had the only non-100% first-pass rate among the five. This is consistent with Stage 3 pool ranking becoming more discriminating as the candidate pool grows: the Dead Reckoning pool contained 45 candidates from which 19 were selected, giving the ranking stage more signal to discard weak candidates before authoring began.

Second, pass rate does not correlate with domain type. Fictional worlds (Grand Larceny, Ironfall, Wavelength, Dead Reckoning) performed at least as well as the real-game-knowledge domain (Age of Empires), despite fictional scenarios requiring the pipeline to generate domain facts rather than retrieve them. The world-lock mechanism (§3) appears sufficient to anchor the generation space for fictional domains.

Third, the pattern of which puzzles failed is informative. All three redesigns in the live-tested scenarios occurred at Stage 8 (blind testing and integration), not at earlier stages. Zero failures were introduced after Stage 8 (no puzzle that passed Stage 8 subsequently failed Stage 9 or later on puzzle-quality grounds). This suggests the gates are functioning: failures that the pool ranking and editorial review stages did not catch are surfaced by the blind test, and no quality failures survive to the delivery stage.

4.3 Stage Timing

Precise commit timestamps are available for Scenario 1 (Age of Empires), which was executed in a single uninterrupted session on February 27, 2026. Scenarios 3–5 ran in overlapping parallel sessions on the same day, making per-stage wall-clock attribution ambiguous. Figure ?? presents the Scenario 1 timing as a reference baseline.

The total elapsed time of 3 hours 17 minutes for a 6-puzzle hunt (5 feeders + meta) from blank scope to ship-ready package establishes a practical throughput baseline. Several features of this timing warrant comment.

Table 3: Stage-by-stage elapsed time for Scenario 1 (Age of Empires), derived from commit timestamps. Stage 5 was embedded within Stage 6 and is not separately measured. Stage 4 was resolved in the same session as Stage 3 with no intervening commit.

Stage	Name	Commit time	Duration
1	Scope	09:16	—
2	Structure	09:57	41 min
3	Puzzle Pool	10:05–10:17	8–12 min
4	Assignment	10:17	≈0 min
5	Meta Design	(embedded in Stage 6)	—
6	Authoring	11:06	49 min
7	Editorial	11:37	31 min
8	Integration	12:01	24 min
9	Live test	12:28	27 min
10	Complete	12:33	5 min
Total (Stages 1–10)			3h 17m

Stage 6 (Authoring) at 49 minutes is the longest single stage despite producing the most content (five complete puzzle files). This is because Stage 6 dispatches parallel AI agents, one per puzzle, so effective per-puzzle authoring time is 49 minutes / 5 = approximately 10 minutes per puzzle. The parallelism is the primary throughput mechanism for scale: a 15-puzzle hunt at the same per-puzzle rate would add only marginally to Stage 6 wall time.

Stage 7 (Editorial) at 31 minutes is the most human-intensive review stage: a human reads all AI-generated editorial reports and decides on pass/revise/cut for each puzzle before Stage 8 begins. This human dwell time does not compress with parallelism in the same way Stage 6 does.

Stage 2 (Structure) at 41 minutes reflects the combined cost of world design and round structure generation for this scenario. For fictional scenarios (Ironfall, Grand Larceny), Stage 2 is longer because the world-lock document must be generated from scratch rather than adapted from an existing IP.

These timings are single-run estimates from commit timestamps. Variance across scenarios was not formally measured; the overlapping parallel execution of Scenarios 3–5 precludes clean per-stage attribution for those runs.

4.4 Failure Analysis

Table 4 catalogs the five failures observed across all scenarios, including the partial Boardgames scenario. The table codes each failure by stage of detection, failure type, and recovery action.

The five failures divide into four mechanistically distinct categories.

Mechanism failure (failure #1): Puzzle V of Age of Empires required solvers to perform integer arithmetic to derive a resource count, with no in-

Table 4: Complete failure inventory across all pipeline runs. “Stage caught” identifies where the failure was first detected, not where it originated. GL = Grand Larceny.

#	Scenario	Puzzle	Stage	Failure type & Recovery
1	Age of Empires	Puzzle V (Post-Imperial)	8 (Blind test)	Pure computation, no deduction layer; extraction mechanism redesigned
2	Ironfall	P01 (Bestiary)	8 (Integration)	Answer mismatch (USHER→UMBRA); meta coordination failure; extraction redesigned
3	Ironfall	P02 (Forge Guide)	8 (Integration)	Answer ambiguity (DHRY / DRESS / QUELL); extraction redesigned
4	Boardgames	M2 (Settlers)	6 (Authoring)	Resource math broken, no valid extraction path; not shipped (workflow test)
5	Grand Larceny	(delivery)	9 (Delivery)	Missing HTML build artifacts; delivery deferred to Stage 11

intermediate deduction step. The Stage 8 blind test identified this as “pure computation”—the answer was uniquely determined but required calculation rather than insight, violating the puzzle craftsmanship rubric (criterion C3: extraction must reward deduction, not arithmetic). This failure type is in principle detectable at Stage 7 but was missed; Stage 8 provided the corrective catch.

Answer coordination failures (failures #2 and #3): Both Ironfall failures arose from a mismatch between feeder-puzzle answers committed during Stage 6 and the meta puzzle’s answer network committed during Stage 5. P01 produced the answer USHER, but the meta required UMBRA at that slot; P02 produced an ambiguous extraction that could validly yield three different answers. These failures are the most structurally dangerous in the pipeline: they cannot be detected at Stage 3 (pool ranking occurs before meta design) and are detectable at Stage 7 only if the editorial reviewer has access to the finalized META.md. In this run, the editorial review did not cross-reference META.md, leaving the coordination failure to be caught at Stage 8 integration. This is a process gap, and a design recommendation follows directly: Stage 7 editorial review should explicitly load and verify answer consistency against META.md.

Construction error (failure #4): The Boardgames meta puzzle M2 (Settlers of Catan variant) was generated with an internal resource-count inconsistency that left no valid extraction path. This failure was caught at Stage 6 during authoring verification—the earliest possible detection point—before any

editorial or integration investment was made. The scenario was already a partial workflow test and the puzzle was not shipped.

Delivery gap (failure #5): Grand Larceny’s Stage 9 delivery preparation produced a valid puzzle set but did not generate the HTML build artifacts required for the digital dossier format. This is not a puzzle-quality failure but a pipeline process gap: the `/hunt site` skill had not yet been invoked, and the delivery checklist did not catch the missing artifacts. Recovery was straightforward (deferred to Stage 11) but the gap illustrates that delivery-format completeness requires explicit checklist enforcement.

The failure distribution by stage (0 at Stage 3, 1 at Stage 6, 3 at Stage 8, 1 at Stage 9, 0 post-Stage 9) supports the primary claim of this paper: the review gates are functioning. Stage 3 pool ranking discards candidates before authoring investment is made; Stage 7 editorial catches mechanism failures; Stage 8 integration catches coordination failures that Stage 7 missed. No puzzle that passed Stage 8 failed on quality grounds at any subsequent stage.

4.5 Cross-Theme Transferability

A central design goal of the pipeline is transferability across themes and settings: the same skill library and reviewer profile set should run a music-themed hunt and a sci-fi RPG hunt without per-scenario code changes. The evidence supports this claim. All five scenarios share the puzzle hunt format; structural transferability to qualitatively different interactive systems (tabletop campaigns, interactive fiction) remains an open question.

Across the five fully-tested scenarios, the only pipeline artifacts that varied per scenario were the `SCOPE.md` (hunt identity brief), the `world/` data files (domain facts and world canon), and the `ROUNDS.md` (round structure). The skill prompts (`toolkit/skills/`), reviewer profiles (`toolkit/profiles/`), rubric criteria, and gate conditions were invariant across all runs.

This invariance is observable directly from the commit history: 142 commits over 2 calendar days include no per-scenario modifications to any skill file. Each scenario’s `CLAUDE.md` imports from `../../toolkit/` without modification. The toolkit commit messages are domain-agnostic (e.g., `Add delivery skills: /hunt print, /hunt props, /hunt site; Add /hunt world skill`) and apply equally to all scenarios that were initialized after the commit.

The reviewer profile library did grow during the two-day run: the initial 12-profile panel was expanded to 25 profiles within the first two hours, then to 28, and ultimately to 29 profiles by end of day 2. This growth was driven by perceived coverage gaps (e.g., adding a cryptic-crossword specialist when Iron-fall required wordplay-heavy extraction), not by domain-specific customization. The profiles that were added apply across all domains.

4.6 Pipeline Maturity Growth

The pipeline is not static. The git history records a clear maturation arc across the two-day scenario run: deficiencies identified in early scenarios drove im-

provements that benefited later scenarios.

The skill library grew from 0 named skills at initialization to 13 skills by the end of the second day. Key inflection points include the addition of `/hunt world` (commit 2534b38, 15:20 on day 1), which was motivated by Ironfall’s requirement for a fictional universe design tool not present in the initial toolkit; `/hunt print`, `/hunt props`, and `/hunt site` (commit 735b379, 15:58 on day 1), added when scenarios requiring physical and digital delivery revealed gaps in the delivery stage; and `/hunt resume` and `/hunt publish` (commit 3f1936e, 22:46 on day 1), added to support scenario resumption across sessions and clean distribution packaging.

Bug fixes also propagated forward: two bugs identified in Age of Empires (skills not auto-updating the scenario `CLAUDE.md` on initialization) were fixed before the Boardgames and Ironfall runs, contributing to the cleaner first-pass behavior observed in the later scenarios.

This maturation pattern is a normal property of pipeline-style system development and is worth naming explicitly: the pass rates reported in §4.2 reflect the pipeline in a state of active improvement, not a frozen system. The Age of Empires scenario (first run, 83% first-pass) and the Dead Reckoning scenario (last run, 100% verified) bookend the improvement arc. Whether this improvement is attributable to pipeline refinements, to the larger and better-ranked candidate pools in later scenarios, or to accumulated experience within the session is a confound the current study cannot fully disentangle; separating these effects is a direction for future controlled evaluation. Practitioners deploying the pipeline for the first time should expect first-run performance closer to the 83% observed in Scenario 1 than the 90.4% aggregate, which blends early and mature runs.

5 Discussion

5.1 Why Stage 3 and Stage 7 Are Highest-Leverage

The asymmetry of failure costs is the central design insight of the pipeline. A puzzle concept that is rejected at Stage 3 (pool ranking) costs only the work of writing a brief — typically a few minutes of prompting and review. A puzzle that clears Stage 3 but fails at or after Stage 7 (editorial review) wastes the full authoring cycle: puzzle construction, answer confirmation, extraction mechanism design, and any associated world-building, amounting to 30–60 minutes of invested effort per puzzle. Two review gates rather than one is therefore not an aesthetic choice but an economic one: the first gate is cheap insurance against expensive downstream work; the second gate is the last high-signal checkpoint before integration costs compound further.

The empirical data is consistent with this framing. Across the 52 puzzles in the five core scenarios, zero failures were detected at Stage 3. This does not mean Stage 3 catches nothing — it means the pool-ranking pass correctly elevated strong candidates and suppressed weak ones before any authoring in-

vestment was made. The three constructive failures (Age of Empires Puzzle V, Ironfall P01, Ironfall P02) emerged at Stage 8 (testing and integration), after Stage 7 editorial review had passed them. These failures are attributable not to editorial blindness but to failure modes that only become visible during answer-path verification: a pure-computation mechanism with no deductive layer, and two answer-coordination mismatches arising from brief-to-meta handoff errors. Both are classes of failure that Stage 7 rubrics can be extended to detect, and both are recoverable at Stage 8 with moderate cost. The remaining two failures were a construction error caught at Stage 6 and a delivery-format gap at Stage 9 — neither attributable to editorial gate failure.

The key operational recommendation from this analysis is direct: do not commit to authoring until Stage 3 ranking is complete, and do not commit to integration until Stage 7 editorial verdict is in hand. At any production scale, the cost savings per filtered puzzle are material.

5.2 The World Lock as a Coherence Mechanism

Stage 2 produces a canonical world document — `LOCKED.md` in the pipeline’s artifact convention — which is treated as read-only by all subsequent stages. This single-document, single-commit mechanism is the pipeline’s primary defense against generation drift: the gradual incoherence that accumulates when later-stage content is generated against a world that has itself been revised.

Without a world lock, the failure mode is subtle and insidious. A character name changed in a late Stage 6 revision would invalidate puzzle flavor text written in Stage 4; a geography change would break a Stage 5 meta mechanism that depended on location names as answer components. These failures are difficult to detect in isolation because each individual artifact may be internally consistent, while the collection as a whole is not. In a multi-stage generative pipeline operating across parallel authoring sessions — as is the case in Stages 5 and 6 of this pipeline — incoherence accumulates silently across agent contexts.

None of the five core scenarios experienced coherence failures attributable to world drift. This negative result is, we argue, precisely the point: the lock worked as intended and the failure class it prevents did not occur. The cost of the lock is negligible — it requires nothing beyond what a scope and structure stage would produce anyway, formalized as a committed artifact. The return is immunity to an entire failure class.

For practitioners adapting the pipeline to new domains, the implication is clear: produce the world lock before running any generative stage, commit it to version control, and communicate its read-only status explicitly in the scenario’s `CLAUDE.md`. The one-time setup cost is dominated by the coherence insurance it provides.

5.3 The Autonomy Gradient

A counterintuitive finding from the autonomy coding analysis (Section 4) is that the pipeline’s middle stages — authoring (Stage 6) and editorial review (Stage 7)

— are where AI operates most autonomously, while the earliest and latest stages involve the greatest proportion of human judgment. Stage 6 in particular was executed entirely by AI agents across all scenarios, with parallel agent launches and no human editing of puzzle files during construction. Stage 7 editorial review was AI-panel-driven in all cases, with human involvement limited to disposition decisions on borderline verdicts.

This gradient is counterintuitive because creative work is typically assumed to require human oversight precisely at the point of creation. The explanation is structural: the middle stages are where domain knowledge is densest and most tractable for AI systems. Stage 6 operates against a fully specified brief (from Stage 4), a locked world (from Stage 2), and a known answer (from Stage 5). The generation task is highly constrained, and the quality rubric is explicit. Stage 7 applies a formalized editorial rubric against a completed artifact — a task well-suited to multi-profile panel evaluation. Constraint density, not creativity, is what drives AI autonomy.

By contrast, Stage 1 (scope definition) requires human judgment about intent — what the hunt should *mean*, what experience it should produce, who the audience is. Stage 11 (polish) requires judgment about edge cases the system did not anticipate: hint calibration for specific solver populations, accessibility accommodations, last-mile delivery decisions. Neither stage admits the kind of structured task decomposition that makes AI autonomous execution reliable. Human involvement is most valuable at the boundaries where structure is thinnest.

This gradient has a practical implication for pipeline adoption: practitioners should expect to invest human time at Stage 1 and Stage 11, and should trust AI autonomy most heavily in the middle. Attempts to increase human involvement in Stage 6 authoring, perhaps out of skepticism about AI-generated puzzle content, are likely to add cost without proportionate quality benefit.

5.4 Limitations

Several limitations constrain the generalizability of these findings.

Single toolkit system. All five scenarios were executed using the same CLAUDE.md-driven skill library and reviewer profile set. Cross-pipeline transferability — whether a different authoring system with different underlying LLM prompts would yield comparable stage-level pass rates — is not demonstrated here. The pipeline *specification* is transferable; the empirical results are specific to this implementation.

AI panel as quality proxy. Pass rates throughout this paper are assessed by AI expert panel review, not by human play-testing. Whether AI panel verdict at Stage 7 correlates with solver enjoyment, puzzle fairness, or aha-moment quality is studied in a companion paper [?]. The results reported here establish pipeline behavior under the AI-panel quality criterion; they should not be taken as evidence about human solver experience without that validation. All quality metrics are derived from AI panel review; correlation with solver enjoyment, engagement, and perceived difficulty is studied in companion work [?]. The

pipeline optimizes for panel-pass quality, which is a proxy for but not identical to player experience.

Maturation confound. The 90.4% aggregate pass rate spans a materially evolving pipeline; improvement attributable to pipeline refinement, larger candidate pools, and accumulated session experience cannot be disentangled.

Absence of baseline condition. We do not report a gate-disabled baseline condition; the causal attribution of quality improvement to pipeline structure is inferential rather than experimentally established.

Scale ceiling. The largest scenario in the empirical set is Dead Reckoning at 19 puzzles (15 feeders, 4 metas). MIT Mystery Hunt scale — 200 or more puzzles with nested meta structure and cross-round interactions — is entirely untested. Pipeline behavior at that scale, particularly the coherence properties of the world lock and the compounding effects of Stage 8 answer-coordination failures, is an open empirical question.

Timing as proxy. Stage timing data is derived from commit timestamps, which capture wall-clock intervals between artifact commits rather than active human-minutes per stage. Stages executed in overlapping sessions (Scenarios 3 through 6 ran in parallel contexts on Day 1) compound this measurement limitation. The timing analysis presented in Section 4 should be understood as an order-of-magnitude characterization, not a precise human-effort accounting.

5.5 Design Recommendations for Practitioners

Five recommendations follow from the empirical analysis.

Run Stage 3 pool ranking before committing to authoring. The cost savings per filtered puzzle are significant at any scale. A pool of 15 candidates ranked by an AI panel requires less than 20 minutes (as observed in Stage 3 of the Age of Empires scenario at 8–20 minutes elapsed); committing to authoring without ranking exposes all candidate slots to the full authoring cost regardless of quality.

Invest in the world-lock document before running generative stages. The lock prevents coherence failures at no additional cost beyond what the scope and structure stages produce in the normal course. The one-time overhead of committing `LOCKED.md` to version control is dominated by the coherence insurance it provides across all subsequent stages.

Run AI expert panels on every puzzle candidate, not only selected finalists. Panel evaluation is cheap enough — and reliable enough relative to the cost of a wasted authoring cycle — that broad application is warranted. Reserving panel review for a pre-selected shortlist defeats the purpose of the Stage 3 gate.

Configure per-scenario scope and world documents; reuse the skill library unchanged. The five-scenario empirical record confirms that the `CLAUDE.md` skill library transfers without modification across domains, scales, and delivery formats. Per-scenario effort should be concentrated in Stage 1 (scope) and Stage 2 (world design), not in pipeline reconfiguration.

Expect and plan for Stage 8 failures. Three of the five pipeline failures surfaced at Stage 8, after editorial review had passed the puzzles. A small Stage 8 failure budget (10% of final puzzle count) should be assumed when estimating delivery timelines, and the pipeline should be operated with sufficient puzzle-pool depth to absorb redesigns without missing the final count target.

6 Conclusion

We have presented an empirical characterization of an 11-stage AI-assisted pipeline for puzzle hunt creation, executed across five independent scenarios spanning game-knowledge, noir fiction, science-fiction RPG, music, and investigation domains. The pipeline produced 52 puzzles across these scenarios, of which 47 passed editorial review on the first pass — a 90.4% first-pass pass rate — with all five failures recoverable by Stage 11. To our knowledge, this is the first cross-scenario empirical characterization of a structured generative pipeline for multi-artifact creative systems.

Three findings from this study are robust across the full scenario set. First, the 90.4% first-pass pass rate, held consistently across five domains and two scales (5–19 puzzles), establishes that the pipeline produces shippable output as a reliable baseline rather than as an occasional outcome. Unconstrained generation of creative artifacts at this level of quality is not achievable without the stage structure the pipeline imposes; the structure is the intervention, not the individual prompts. Second, the concentration of failures at Stage 8 (testing and integration), rather than at the review gates (Stages 3 and 7), confirms that the gates are functioning as designed. A gate that catches everything would leave nothing for downstream stages to fail; a gate that catches nothing would redistribute failures uniformly across stages. The empirical distribution — zero gate-stage failures, three Stage 8 failures — is consistent with gates operating at the right threshold: permissive enough to let strong candidates through, selective enough to filter weak ones before authoring investment. Third, the same skill library — 13 CLAUDE.md-driven skills and a 29-profile reviewer panel — executed all five scenarios without per-scenario modification. Only the scope and world documents varied. This confirms cross-theme, cross-setting transferability at the specification level, which is the form of transferability that matters for practical adoption.

For practitioners considering pipeline adoption, the central message is that structure is the primary quality lever. Individual prompt quality, reviewer profile sophistication, and domain-specific knowledge all contribute, but the two review gates at Stages 3 and 7 account for the majority of the quality guarantee. The world lock — a single committed document that all subsequent stages treat as read-only — provides coherence insurance at negligible cost. These are structural decisions, not implementation details, and they are the first thing a practitioner should get right.

Several questions remain open. Pipeline behavior at MIT Mystery Hunt scale — 50 or more puzzles with nested meta structure and cross-round answer

dependencies — is the most pressing empirical gap; the coherence properties of the world lock and the compounding probability of Stage 8 coordination failures both warrant investigation at that scale. The correlation between AI panel editorial verdict and human solver experience is studied in a companion paper [?]; integrating those results with the stage-level pass rates reported here is a natural next step. Cross-setting pipeline adaptation — to tabletop campaign generation, interactive fiction, or educational puzzle design — would test whether the specification-level transferability observed across five puzzle hunt scenarios spanning different themes and settings extends to structurally different creative artifact classes. Finally, the failure data collected across these five scenarios provides a natural training signal for automated skill refinement: failure types can be mapped to the stage and rubric criterion that should have caught them earlier, enabling systematic improvement of the review gates over time.

The full pipeline specification, including the `CLAUDE.md` skill library, reviewer profiles, and scenario artifacts for all five scenarios, is available at [URL]. We offer it as a reusable foundation for researchers and practitioners working on structured generative pipelines for complex creative systems.

References